Project ID : R-26-DS-011
Student ID : IT22129758
Name : Edirisinghe E.A.K.
Individual Component : Multilingual emergency conversational AI agent for rapid clarification

2026/01/01 : Identified the general research domain and discussed the real-world importance of an AI-based emergency call and SMS triage system for disaster response in Sri Lanka, especially for low-resource local languges such as Sinhala and Tamil.

2026/01/04 : Researched how emergency communication systems work and focused on the challeges in processing multilingual emegency calls and SMS messages in Sinhala, Tamil and mixed language inputs.

2026/01/07 : Collected related research papers and reviewed previous studies connected to speech, text and emegency message analysis.

2026/01/08 : Visited DMC for the first time and discussed our research topic. We talked about the importance of this research to DMC and discussed possible ways to collect data for our study

2026/01/10 : Prepared my part of the TAF document and focused on how the system can analyze emergency conversational AI agents.

2026/01/12 : Submitted the TAF document.

Student progressing well in research project Yes / No
Signature :

Research Project with log entity form

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2025/12/03 : Meet our research project supervisor and disscuss about few topics to do our research

2025/12/08 : Meet our research project supervisor and discuss about few topics to do our research

2025/12/16 : Finalized our research topic named" AI based emegency call and sms triage system for disaster response in Sri Lankan low resource local languages" as a research topic

2025/12/18 : Meet our research supervisor and discussed our components with other member.

2025/12/20 : Finalized research topics with components.

2025/12/21 : Register our research group and topic with relevant components.

Student progressing well in research project yes/No

Signature :

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2026/02/04 : Started to identify what data should be collected for my function and discussed the main inputs needed for the emagency classification component.

2026/02/07 : Attended a meeting and discussed the function with industry-related experts. Gained useful ideas about emergency communicaton - language challenges, and possible user inputs.

2026/02/12 : Prepared the external data collection request form and submitted it to CDAP for data collection purposes.

2026/02/16 : Arranged a meeting with group members and discussed each member's user inputs and outputs to keep the overall system connected properly.

2026/02/20 : Arranged a meeting with group members and supervisor and presented our user inputs and outputs. After the discussion, made several changes to improve the function.

Student progressing well in research project Yes / No

Signature :

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2026/03/01 : Started preparing the research proposal presentation and organized the main ideas related to the project topic.

2026/03/05 : Identified the structure of the individual report and presentation and selected the content related to my component.

2026/03/07 : Collected the external data collection letter form CDAP and visited DNC to hand over the letter for data collection purposes.

2026/03/14 : Finalized the research proposal report with the relevant details of the proposed system and Individual component.

2026/08/22 : Arranged a meeting with group members and supervisor to discuss each member's report and recieve feedback for improvement.

2026/03/24 : Finalized the research proposal presentation and obtained supervisor signatures for the research proposal report.

2026/03/26 : Presented the research proposal presentation to the panel.

Student progressing well in research project Yes / No
Signature:

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2026/04/01 : Arranged a meeting with supervisors and group members to discuss the research topic.

2026/04/04 : Searched for suitable models to use in my research component and gained additional knowledge about AI models related to the function.

2026/04/09 : Started collecting data for the research.

Student progressing well in research project Yes/ No
Signature :

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2026/04/11: Started data collection part of my function and identified what data I need.

2026/04/16: Set up Cloud speech Service resource in to test speech-to-text functionality for the multingual emergency call project

2026/04/20: Had a team meeting to discuss the project plan and devide tasks among members.

2026/04/23: Started Speech to text, the next part is extract part.

2026/04/26: Had a team meeting to discuss the project plan and devide tasks among members.

Signature.

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2026/05/01 : Implemented the speech to text module. Tested audio transcription and saved the transcript output as a json file.

2026/05/03 : Created the NER dataset in CoNLL format. Each word was labeled using BIO labels such as B- INCIDENT, I- INCIDENT

2026/05/05 : Trained and Tested Information extraction models such as CRF, BiLSTM-CRF and XLM. Compared models using key metrics.

2026/05/07 : Implemented off-topic detection. Created a dataset with text and label columns and trained Logistic Regression, SVM and Naive Bayes.

2026/05/09 : Implomed missing field detection and clarification question generation.

2026/05/10 : Had a team meeting to discuss the project plan and devide tasks among members

2026/05/11 : Present the PPI to the Panel.

Singnature :

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Name : E.A.K. Edirisinghe
Individual Component : Multilingual emergency Conversational AI agent for rapid clarification

2026/05/13 : Conducted a team meeting to discuss individual component progress and project timeline.

2026/05/15 : Started correcting the mistakes highlighted by the evaluation panel during the review.

2026/05/17 : Improved model accuracy by testing different models and adjusting training methods.

2026/05/20 : Explored a hybrid multi-agent approach for clarification question generation and validation.

2026/05/24 : Researched Twilio integration methods for connecting real-time emergency calls to the backend.

2026/05/26 : Conducted another team meeting to review completed tasks and plan remaining implementation work.

2026/05/28 : Finalized the working pipeline to process audio, extract information, detect missing fields and generate question.

Singnature :

Project ID → R-26-DS-011
Student ID → IT20129758
Name → E A. K. Edirisinghe
Individual Component → Multilingual emergency conversational AI agent for rapid clarification

2026/06/02 : Ran the routing evaluation script on the gold dataset (146 examples) to measure the baseline route accuracy of the current system.

2026/06/07 : Added Tamil relief-request cues and common relief items (rations, blankets, mats) so that welfare requests from Tamil callers are no longer misrouted as off-topic.

2026/06/08 : Designed the narrator-perspective module that detects when a caller is reporting on behalf of a third party and planned how questions should change based on that.

2026/06/15 : Improved the logic for deciding whether a location is specific enough to dispatch to (for example schools, ~~hosp~~ hospitals, and campuses), instead of asking for a road or house number again.

2026/06/22 : Corrected Sinhala number-word handling so compound and irregular numbers are read correctly.

2026/06/28 : Strengthened the emergency triage check so that a single serious symptom is recognised as high risk and re-ran the evaluation to confirm the emergency veto reached 100%.

Signature :

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2026/07/08 : Added the wellness-check response for a confused or disoriented caller, so the system asks whether the caller is safe before giving a standard emergency-only reply.

2026/07/08 : Added the dispatch follow-up response, so a caller who has already reported an emergency and asks when help will arrive gets reassurance instead of a redirect.

2026/07/14 : Wrote the rules that decide when a wellness check should be used, so it is asked only once per call and not repeated.
2026/07/17 : Had a team meeting.
2026/07/20 : Tested the wellness check and follow-up responses on single turns and recorded the results.

2026/07/24 : Tested the off-topic gate on the live panel with different caller senarios and recorded the wrong responses found during the session.

2026/07/29 : Reviewed the off-topic evaluation numbers and listed the remaining misclassified examples for the next round of changes

Signatures :

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2026/08/04 : Reworked the off-topic wording so each situation (off-topic, ambiguous, follow-up) uses a clear and separate question, and checked the new wording on the live panel.

2026/08/10 : Updated the Tamil and Singlish versions of the same wording, keeping the language mixing consistent with each language type.

2026/08/18 : Started the people-count mechanism for vague answers such as "many people", with a limit on how many times the system asks for an exact number.

2026/08/22 : Added a categorical quantity list (many, a few, several, a crowd) and a rule for an explicit "I cannot count" reply.

2026/08/29 : Ran the live people-count scenarios and found that the system still repeats the question in most cases; traced the cause to the order of the decision steps.

2026/08/31 : Recorded the people-count failures in the evaluation notes so they can be compared after the next change.

~~Singnds~~ Signatures:

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2026/09/03 : Ran the routing evaluation again after the off-topic wording changes to confirm that route accuracy and the emergency veto did not drop.

2026/09/07 : Checked the narrator-perspective module on controlled examples (third-party reports and self-reports) to confirm that self-reports were not affected by the change.

2026/09/11 : Checked the location sufficiency rules on campus, hospital and two-area example, and confirmed that a bare area name is still treated as too vague.

2026/09/16 : Re-tested the Sinhala number handling on the compound and irregular forms and confirmed that the built-in numerical test cases still pass.

2026/09/21 . Had a team meeting.

2026/09/25 : Ran the five people-count scenarios again after the state changes and recorded that the people question is still not resolved in most of them.

Signature :

Project ID : R-26-DS-011
Student ID : IT22109758
Name : E.A.K. Edirisinghe,
Individual component : Multilingual emergency conversational AI agent for rapid clarification.

2026/10/01 : Documented the design decision and iteration log for the component for the thesis.

2026/10/03 : Wrote down the next steps for the people-count problem

2026/10/04 : ~~Downloaded~~ Reviewed the thesis draft chapter by chapter against the department's guidelines and listed the sections that still need more details.

2026/10/05 : Updated the component chapters with the lastest code changes and the iteration log.

2026/10/07: Checked the thesis figure and the in-text citations against the reference list, and marked the references that still need verification.

2026/10/08 : Worked on the results chapter, adding the routing and numerical results and recording the people-count result as not yet resolved.

2026/10/09 : Continued the theses draft and listed the remaining work before the final version.

Signature :